

Name:- Srishti Kumari

Artificial Intelligence InternsElite

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Traffic Sign Recognition System using CNN

1. Introduction

Traffic Sign Recognition (TSR) is an important application of computer vision and deep learning used in autonomous driving and driver assistance systems. The goal of this project is to develop a model that can accurately classify traffic signs from images.

In this project, we used the **German Traffic Sign Recognition Benchmark (GTSRB)** dataset and implemented a Convolutional Neural Network (CNN) model to classify traffic signs into 43 different categories.

2. Objective

- To build an image classification model using deep learning
- To recognize and classify traffic signs
- To understand the working of Convolutional Neural Networks (CNN)
- To achieve good accuracy using optimized dataset and training

3. Dataset Description

The dataset used is **GTSRB (German Traffic Sign Recognition Benchmark)**, which contains images of traffic signs belonging to 43 different classes.

Dataset Components:

- **Train Folder** – Contains labeled training images
- **Test Folder** – Contains testing images

- **Meta.csv** – Contains metadata of traffic signs
- **Train.csv & Test.csv** – Contain image paths and labels

Dataset Modification:

Due to time constraints, **5000 images** were randomly selected from the training dataset for faster processing.

4. Methodology

Step 1: Data Loading

- Dataset uploaded to Google Drive
- Accessed using Google Colab
- Images loaded using OpenCV

Step 2: Data Preprocessing

- Images resized to **32×32 pixels**
- Normalization performed (pixel values scaled between 0–1)
- Labels converted using **One-Hot Encoding**

Step 3: Data Splitting

- 80% data used for training
- 20% used for validation

Step 4: Model Building (CNN)

A Convolutional Neural Network was used with the following layers:

- Convolution Layer (32 filters)
- Max Pooling Layer
- Convolution Layer (64 filters)

- Max Pooling Layer
- Flatten Layer
- Dense Layer (128 neurons)
- Dropout Layer (to prevent overfitting)
- Output Layer (43 classes using Softmax)

Step 5: Model Training

- Optimizer: Adam
- Loss Function: Categorical Crossentropy
- Epochs: 5
- Batch Size: 32

Step 6: Model Evaluation

- Model tested on test dataset
- Accuracy calculated

5. Results

- Training completed successfully
- Model achieved approximately **69–70% accuracy**
- Performance is satisfactory considering reduced dataset size

6. Tools & Technologies Used

- Python
- Google Colab

- TensorFlow / Keras
- OpenCV
- NumPy, Pandas
- Matplotlib

7. Advantages

- Efficient traffic sign detection
- Useful in autonomous vehicles
- Reduces human error in driving

8. Limitations

- Reduced dataset size affects accuracy
- Some images failed to load due to path issues
- Model can be improved with more training

9. Future Scope

- Use full dataset for higher accuracy
- Apply data augmentation
- Implement real-time detection using webcam
- Deploy model using web or mobile applications

10. Conclusion

This project successfully demonstrates the use of deep learning in image classification. A CNN-based model was developed to classify traffic signs with reasonable accuracy. The project highlights the importance of preprocessing, model selection, and optimization in machine learning workflows.

11. References

- [GTSRB Dataset \(Kaggle\)](#)
- [TensorFlow Documentation](#)
- [Keras Documentation](#)
- [OpenCV Documentation](#)